

Measurement of the Mass and Width of the Z-boson Using the LHCb Detector – Final Report

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The Z-boson mass and width were measured using the LHCb detector using $z \rightarrow \mu\mu$ decays. This was the first dedicated Z measurement using data from the LHC run 3 set which used a centre of mass energy of 13.6 TeV and an integrated luminosity of 1150 pb⁻¹. The $\Upsilon(1S) \rightarrow \mu\mu$ decay channel was used for calibration, resulting in a Z-boson mass and width measurement,

$$M_Z = 91.119 \pm 0.011 \pm 0.004 \text{ GeV},$$

$$\Gamma_Z = 1.90 \pm 0.03 \pm 0.12 \text{ GeV},$$

and correlation 0.065 where the first error is statistical and the second is systematic. These results disagree with previous measurements and predictions from the standard model.

I. Introduction

Particle physics theory is well described by the standard model [1]. This is a quantum field theory that through a Lagrangian, explains the existence of the fundamental particles and their interactions. The standard model has successfully led to the predictions of the top quark [2, 3], tau neutrino [4] and Higgs boson [5]. However, this theory is incomplete as there are many phenomena it cannot explain such as neutrino mass and oscillations as well the matter/anti-matter asymmetry observed in nature. The main focus of modern particle physics is performing rigorous tests on this theory to find physics beyond the standard model. One of the particles in the standard model is the Z-boson which is the electrically neutral mediator for the weak interaction, responsible for processes such as beta decay and neutrino interactions.

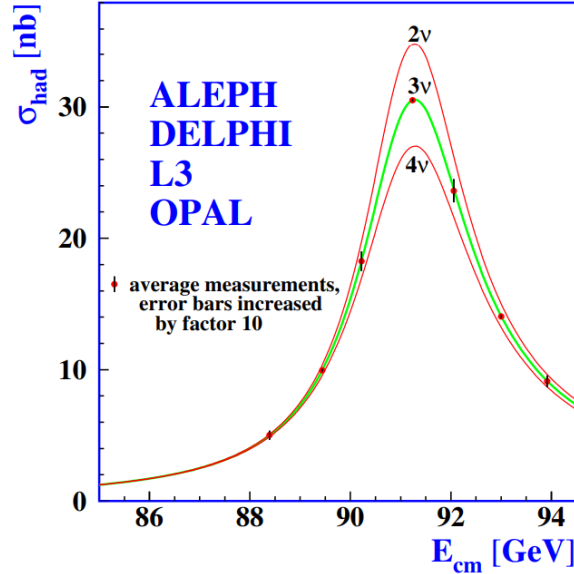


Figure 1: Measurements of the hadronic cross-section at energies around the Z mass. These assume the only invisible decay products of the Z are neutrinos and that they have standard model couplings [6].

The measurement of the mass and width presented in this report is in an attempt to compare with previous measurements and standard model predictions. Accurate measurements of the Z-boson mass have helped constrain other parameters such as the weak mixing angle and Higgs mass prior to its discovery [6]. The Higgs boson, largely seen as the biggest discovery in modern particle physics, was theorised to explain how the W and Z-bosons acquired mass, therefore understanding the Z mass was very important for this. The mass of the Z-boson also characterises the strength of the weak force [7]. Since the Z-boson decays into fermion anti-fermion pairs, its width helps determine the number of generations for fermions. For example, LEP used the observed hadronic cross-section to compare to the standard model predictions for various generations of neutrinos as seen in Fig. 1. This provides strong experimental evidence for the existence of exactly 3 generations of neutrinos.

Many experiments have performed their own analyses of the mass of the Z-boson with the most accurate and well known being the measurement from LEP [6]. Other measurements have been made by CDF [8], CMS [9] and most recently LHCb [10]. The previous LHCb measurement used data from run 2 at the LHC, whereas this project uses more recent run 3 data using the upgraded LHCb detector [11] and also aims to measure the Z-boson width, in addition to the mass. As is convention in high energy physics, natural units of $\hbar = c = 1$ have been used throughout this report.

II. Theory

A. Brief history of electroweak theory

Electroweak theory began in 1930 with Pauli's proposal of "neutrons" as a way to conserve energy in β -decays of Li-6 nuclei and explain the observed continuous spectrum. He proposed, in a letter, that there could exist a neutral spin-1/2 particle that had mass of the same order of magnitude as an electron [12]. After Chadwick discovered the particle now commonly known as the neutron in 1932, Pauli's particle was changed to neutrino [13]. Using this new discovery, Fermi was able to complete the first model for the weak interaction [14]. This version of the weak interaction described β -decay using four fermionic fields (proton, electron, neutron and neutrino) with a constant coupling strength, analogous to how quantum electrodynamics (QED) models the radiative emission of light quanta from an atom [15]. This model succeeded in explaining the continuous β -decay but was found to be incomplete in describing the interaction between protons and neutrons within the atom [16]. Another weakness of this theory was that loop calculations diverged, causing the theory to be non-renormalisable. This meant the theory failed at higher energies.

In 1967, the Wu experiment [17] showed that the weak interaction violates parity through measuring the spins of emitted electrons of Co-60 nuclei as per previous predictions [18]. It had already been theorised that if you impose local gauge symmetry, gauge bosons appear naturally. Using these ideas and Fermi's theory, Glashow, Salam and Weinberg unified the electromagnetic and weak interaction into a single electroweak theory that brought together the previously separate photon and W-boson with a predicted Z-boson [19, 20, 21] as the mixing of the gauge fields that arise from $SU(2) \times U(1)$ symmetry. In this new theory, the constant coupling strength is replaced with a strength proportional to the mass of one of these exchange bosons, fixing the divergence issue of Fermi's theory.

B. The Z-boson

Gauge bosons that are introduced in this way were supposed to be massless. This problem was solved via electroweak symmetry breaking; the theory that due to the non-zero vacuum expectation value of the Higgs potential, the $SU(2) \times U(1)$ is broken for the vacuum and the bosons acquire mass. The gauge fields of this symmetry are labelled W^1, W^2, W^3 and B with the first three coming from the $SU(2)$ group and the final com-

ing from $U(1)$. The W^1 and W^2 fields mix to form the W^+ and W^- and the W^3 and B fields mix to form the Z -boson and the photon. Under this mixing the Z -boson field forms as $Z = \cos \theta_W W^3 - \sin \theta_W B$ [22], where θ_W is the weak mixing angle, defined as $\tan \theta_W = \frac{g_1}{g_2}$ for the gauge coupling constants of $SU(2)$ and $U(1)$, g_1 and g_2 . From this term's interaction with the Lagrangian, at tree level, the predicted mass of the Z -boson is,

$$M_Z = \frac{v}{2} \sqrt{g_1^2 + g_2^2}, \quad (1)$$

where v is the Higgs vacuum expectation energy. This results in a standard model prediction of $m_Z = 91.205 \pm 0.009$ GeV when fit with global electroweak data [23].

The width of the Z -boson is calculated from the same gauge theory but is instead calculated as a sum of smaller decay widths via each decay channel. Each $Z \rightarrow f\bar{f}$ contributes $\Gamma_f = \frac{G_F M_Z^3}{6\sqrt{2}} N_c^f$ to first order where N_c^f is the number of colour charge options for fermion f [24]. This means the total Z width is,

$$\Gamma_Z = \sum_f \frac{G_F M_Z^3}{6\sqrt{2}} N_c^f \quad (2)$$

From the same fit used for the mass above, the standard model prediction for the Z -boson width is $\Gamma_Z = 2.4942 \pm 0.0007$ GeV [23].

C. Experimental measurements of the Z

The first experimental evidence for the Z -boson came from the Gargamelle collaboration through the discovery of hadron-neutrino interactions that did not have a charged lepton in the final state [25]. This could not be explained by the charged weak interaction, so a weak neutral current must exist. Due to the large mass of the Z -boson (approximately 91 GeV) its discovery was hindered until colliders were developed with $\sqrt{s}$ great enough for its production. This came in the form of the Super Proton Synchrotron at CERN, which provided the first experimental observations for both the W and Z -bosons in 1983 by the UA1 experiment and later the UA2. These experiments gave initial measurements of the Z -boson to be $m_Z = 95.2 \pm 2.5 \pm 3.0$ GeV for UA1 and $m_Z = 91.9 \pm 1.3 \pm 1.4$ GeV for UA2 where the first error is statistical and the second error is systematic [26].

There have been multiple experiments since that have measured the Z -boson mass, most recently in a dedicated measurement by LHCb [10], as well as indirect measurements from CDF [8] and CMS [9] as calibration for their W mass analysis. It is only possible to

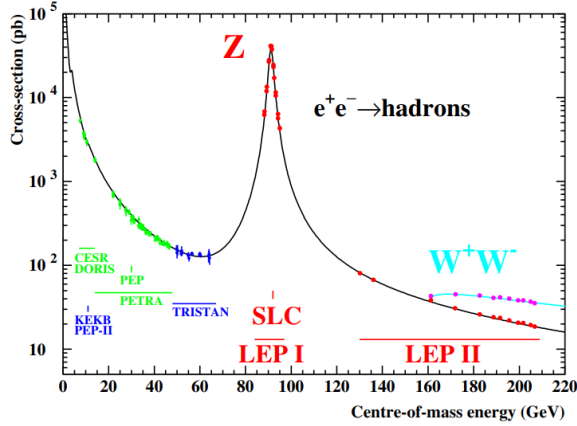


Figure 2: The hadronic cross-section of the $e^+e^- \rightarrow f\bar{f}$ events as measured by LEP [6]. The solid line is the standard model prediction and the points are the experimental data. Various energy scales of e^+e^- accelerators are shown for comparison.

use the Z mass as a calibration because of the precision of the leading measurement by LEP [6]. The LEP was an e^+e^- collider which was approximately 27 km in circumference and consisted of four interaction regions, each with their own detector and experiment [27, 28, 29, 30]. At its peak, LEP produced approximately 1000 Z events per hour across the four experiments [31]. The process $e^+e^- \rightarrow f\bar{f}$ has two decay channels; via a photon or a Z-boson. Fig. 2 shows how these channels interact in the hadronic cross-section. This features a $1/s$ fall off from virtual photon exchange and a resonance peak at the Z mass. This is because photons are massless and have no preferred energy scale but the Z-bosons are massive and require energy equal to their mass to form. This resulted in a measurement for the Z mass and width of $m_Z = 91.1875 \pm 0.0021$ GeV and $\Gamma_Z = 2.4952 \pm 0.0023$ GeV from the peak and shape of this resonance. The main source of uncertainty in this experiment was the accuracy of the beam energies. By using the precession of the spin of polarised particles, beam energies were measured to within 2 MeV [6], this systematic error could be minimised, leading to an incredibly precise measurement.

D. The LHCb detector

The LHCb detector is a single-arm forward spectrometer, which means it starts at the interaction point and extends in a single direction past it along the beam axis. At this interaction point is a silicon-strip Vertex Locator (VELO) which is used in vertex reconstruction [32]. It also consists of a 4 Tm magnet for measuring momenta and identifies particles using the RICH, ECAL, HCAL and, most importantly for this project, Muon detectors. The muon detector has five stations labelled M1-M5 along the beam axis, each with two separate sides for the ease of maintenance as shown in Fig. 3. The M1 is in

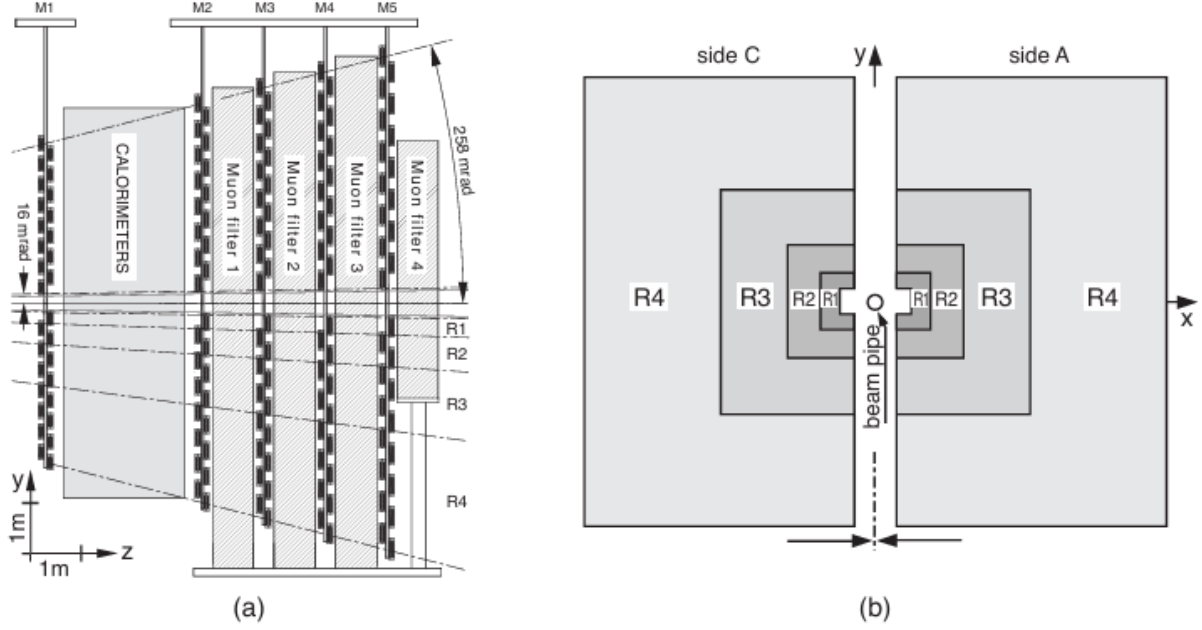


Figure 3: (a) Side-on view of LHCb muon detector. (b) Layout of regions R1-R4 [33].

front of the calorimeter and acts as the Level-0 muon trigger to improve measurements of its transverse momentum, detecting muons with 12 GEM detectors that can handle the high incident radiation. The other stations are after the calorimeter and are separated by 80 cm of iron absorbers that stop non-muon particles and detect muons using highly efficient MWPCs [33]. The hits of these stations are used to form tracks of the muons inside the detector. Within each station there are four regions labelled R1-R4 of increasing distance from beam axis, which allow for variance in detector design within each individual station.

Momenta are measured by the curvature of the charged particle as they travel through the magnetic field, by comparing the Lorentz force with the classical force of circular motion to get a measurement of p/q [34]. This causes a bias as particles will deflect differently based on their charge and small tracking misalignments will overestimate or underestimate the momenta depending on the sign of the particles charge. In an attempt to reduce this bias, the polarity of this magnet is flipped throughout data taking from *MagUp*, where positive particles are deflected towards the centre of LHCb, to *MagDown* [35]. This does not eliminate the bias but there are several methods that can be used for further reduction; for example, the pseudomass method [36].

The LHCb detector is naturally described using coordinates of the pseudorapidity, η and azimuthal angle, ϕ . The pseudorapidity is a function of the polar angle with the LHCb detector covering a range of $2 < \eta < 5$ and is used in detector physics due to its

Lorentz invariance under boosts along the beam axis [37]. The azimuthal angle describes the angle around the beam axis in the transverse plane.

E. Event reconstruction

Candidate $\Upsilon(1S)$ and Z events are reconstructed in the same way. Events that pass all the triggers are stored via a ROOT tree file which is then fed into analysis software that initially interprets the data. The resultant file gives all recorded measurements of the muon pairs in detector coordinates of (p_T, η, ϕ) where p_T is the transverse momenta of the muons. This is convenient for the application of selection cuts to the data used to reduce background, further explained in Sec. III.C.. These measurements could then be converted into Cartesian momenta, (p_x, p_y, p_z) for ease of calculation. When the momenta are in this form, the mass can be reconstructed simply from the relativistic energy momentum relation,

$$E^2 = p^2 + m^2, \quad (3)$$

for a particle's energy, E , if it has mass, m , momentum, p and natural units have been used once again. For both the $\Upsilon(1S) \rightarrow \mu\mu$ and $Z \rightarrow \mu\mu$ reconstructions, Eq. 3 is rearranged for the mass and the input energy and momentum are just the sum of the energies and momenta of the two final state muons. The muon momenta are recorded individually in the decay tree and the muon energies are calculated also using Eq. 3 individually for each muon, using the globally averaged muon measurement for its mass, 0.1057 GeV [38]. The error on this mass is neglected as it is of order 10^{-9} . This will not be the same for every event due to many factors that can affect the measured momentum of the final state muon including ionisation of detector material [39] and photonic emission before observation that can occur in QED [40].

F. Calibration

The LHCb detector response to sources of systematic uncertainty on momentum, p can be modelled using,

$$p \rightarrow p(1 + \alpha)(1 + \frac{\beta}{p})(1 + \delta qp)(1 + \sigma Np)(1 + \sigma' N'), \quad (4)$$

where α , β and δ are parameters for the scaling of the momentum, σ and σ' are parameters for the resolution and N and N' are normally distributed random numbers with a mean of

0 and a standard deviation of 1. The β and δ parameters correspond to the energy loss via ionisation [39] and charge dependent curvature bias [36] effects in the detector. Charge dependent curvature bias arises from small misalignments in the tracking system over-estimating or under-estimating the momentum of a charged particle based on its charge. This is because the measured curvature is broadened or flattened by the error in tracking as mentioned in Sec. II.D. Due to time constraints, for simplicity, these parameters were assumed to be 0 as well as the second smearing parameter σ' since the σ smearing, proportional to $1/p$, is more dominant for the momenta as it is the resolution in curvature.

The parameter α is a momentum offset caused by miscalibration of the bending power of the magnet or tracker length scale and can be determined from fitting the simulation to the data with this as a free parameter [10] or mathematically from propagating this error on observed muon momenta through the muon kinematics. Applying the rearranged Eq. 3 for a particle with two decay products gives $m^2 = (E_1 + E_2)^2 - (\vec{p}_1 + \vec{p}_2)^2$ for decay products 1 and 2. Expanding these out and using Eq. 3 to replace the energy of the products in terms of its mass and momenta gives $m^2 = m_1^2 + m_2^2 - 2(E_1 E_2 - \vec{p}_1 \cdot \vec{p}_2)$. Under the massless approximation of decay products, $m_1 = m_2 = 0$, leaving $m^2 = 2p_1 p_2 (1 - \cos\theta)$ where θ is the angle between the two products. This is a very good approximation since $\frac{m_\mu}{m_Z} \approx 10^{-3}$. Therefore if the observed momenta are mismeasured and require scaling of $1 + \epsilon$, where ϵ is a small parameter, then this propagates through as $m \rightarrow (1 + \epsilon)m$. Since this is a simple linear relation, ϵ can be calculated separately for the simulated and observed data using $\epsilon = \frac{m}{m_{\text{ex}}} - 1$ where m_{ex} is the expected value of the mass. Then given an expected m_{ex} , α can be calculated using,

$$1 + \alpha = \frac{1 + \epsilon_{\text{data}}}{1 + \epsilon_{\text{sim}}}. \quad (5)$$

This scales the mass of simulated events closer to its observed value and reduces the systematic uncertainty on p .

The parameter σ can also be determined from fitting the simulation and data but can also be derived mathematically, similar to α . The aim of this basic momentum smearing is to adjust simulated momenta so its narrow idealised width better matches the observed data. This resolution is not resolution in momentum but rather resolution in curvature because of how the detector measures momentum. Starting from $p = \frac{q}{c}$ where p , q , c are the momentum, charge and curvature respectively and $c = q/p$. Taking the derivative gives $\frac{dp}{dc} = -\frac{q}{c^2}$ which using σ for the width of the variable, $\sigma_p = -\frac{q}{c^2}\sigma_c = -\frac{p^2}{q}\sigma_{q/p}$. This

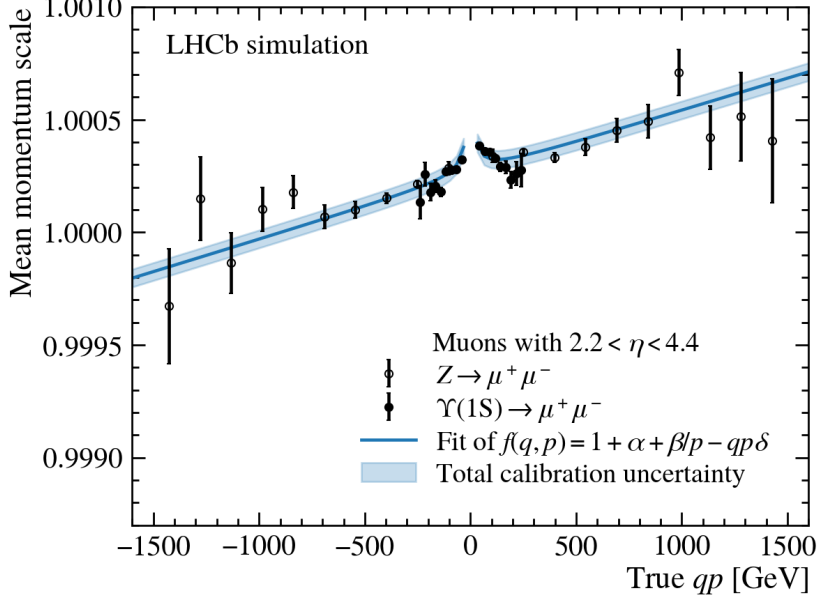


Figure 4: The validity of the linear momentum response in Eq. 4 on calibration for $Z \rightarrow \mu\mu$ events, derived from $\Upsilon(1S) \rightarrow \mu\mu$ decays [41].

gives the relation between the momentum resolution and curvature resolution $\frac{\sigma_p}{p} = \bar{p}\sigma_{q/p}$ since the magnitude of muon charge will be 1 and $\bar{p}$ is a statistically typical momentum of the muons. $\sigma_{q/p}$ is added to adjust the simulated momenta such that $(\sigma_{q/p})_{\text{data}}^2 = (\sigma_{q/p})_{\text{sim}}^2 + (\sigma_{q/p})^2$. From this,

$$\sigma_{q/p}^2 = (\sigma_{q/p})_{\text{data}}^2 - (\sigma_{q/p})_{\text{sim}}^2 = \frac{1}{\bar{p}^2} \left(\left(\frac{\sigma_p}{p} \right)_{\text{data}}^2 - \left(\frac{\sigma_p}{p} \right)_{\text{sim}}^2 \right) \approx \frac{1}{\bar{p}^2} \left(\left(\frac{\sigma_m}{m} \right)_{\text{data}}^2 - \left(\frac{\sigma_m}{m} \right)_{\text{sim}}^2 \right). \quad (6)$$

Here the q/p subscript is added for clarity but is usually neglected as it is in Eq. 4. Since this is a resolution not a simple scaling as with α , it needs to be applied statistically to the simulation, as further explained in Sec. III.D.

Due to the order of magnitude difference between the Z-boson mass and the $\Upsilon(1S)$ mass, the validity of applying calibration from the $\Upsilon(1S)$ to Z had to be tested. The higher mass results in a higher typical momentum of final state muons from a Z decay than from an $\Upsilon(1S)$. The forward design of the LHCb detector helps this by observing $\Upsilon(1S)$ candidates with boosted momentum. Fig. 4 shows how applicable Eq. 4 is for $Z \rightarrow \mu\mu$ reconstruction under the imposed selection cuts. This shows that the model is accurate for muons with low magnitude of momenta but can struggle more for higher momenta muons.

III. Methodology

A. Dataset

Data were taken in 2025 during run 3 of the LHC using the upgraded LHCb detector, corresponding to a centre of mass energy, $\sqrt{s} = 13.6$ TeV and integrated luminosity, $L = 1150 \pm 70$ pb⁻¹. This was split roughly evenly between the two magnet polarities and primary event selection was performed using the trigger system [42, 43]. This results in 4,079,241 $\Upsilon(1S)$ and 205,314 Z candidate events before selection cuts are applied.

B. Simulation Details

Simulated events were generated using PYTHIA [44]; a Monte Carlo event generator used to model particle production in collisions with a specific LHCb configuration [45]. These events can be input into GEANT [46] which can reproduce the interaction of generated events with the LHCb detector. This is an accurate representation of the detector response but cannot perfectly reproduce the real experimental environment. For example, effects such as small misalignments in the tracking system and non-uniformities in the magnetic field can decrease the momentum resolution observed in data. To compensate for this, simulated events require calibration.

C. Selection cuts

To reduce background, selection cuts were made on both $\Upsilon(1S)$ and Z candidates. Muon candidates for both are required to be in the range $2.2 < \eta < 4.4$ which has the added benefit of slightly reducing the effects from the non-uniformity of the detector. Muon candidates also had a cut on their transverse momenta; $p_T > 5$ GeV for $\Upsilon(1S)$ candidate muons and $p_T > 20$ GeV for Z candidates. The cut-off p_T is expectedly lower for the $\Upsilon(1S)$ due to the lower average momenta of its final state muons resulting from its significantly lower mass. These selection cuts did affect the distribution of muon p_T of the $\Upsilon(1S)$ events so that the fitted background was required to be able to increase in slope.

D. Applying the Calibration

In typical analyses; for example, in analyses of W bosons [8, 9], the Z resonance is used as calibration but this is not helpful in a Z analysis. Instead, a decay is required of a particle with similar precision in its mass. Here the $\Upsilon(1S)$ is used, using the specific

decay channel $\Upsilon(1S) \rightarrow \mu\mu$. The $\Upsilon(1S)$ is a $b\bar{b}$ meson with a narrow mass resonance at $9.46040 \pm 0.00009 \pm 0.00004$ GeV where the first error is the statistical uncertainty and the second is the systematic uncertainty [24]. This value is the global average based on measurements across multiple experimental groups from the particle data group. The 1S resonance is used as it is the lowest lying vector $b\bar{b}$ state, making it the easiest to measure. The validity of using the $\Upsilon(1S)$ as a calibration for Z-boson measurements despite its significantly smaller mass was discussed previously in Sec. II.F.

To achieve the calibration parameters in Eq. 5 and Eq. 6, the simulated and observed data were fitted using a fitting function that consisted of a combination of two crystal ball functions [47] with a linear background. Simulated events were manually set to have no background as the simulated data should only consist of $\Upsilon(1S) \rightarrow \mu\mu$ events. Multiple models of the background were tested, but a linear background was found to be the best description of the data. The crystal ball functions were used to model the Gaussian core and the non-Gaussian tails that arise from final state muons emitting photons before detection via QED effects [40]. The two crystal ball functions shared a common mean and tail parameters but had differing widths to describe both the narrow core and broader component from resolution effects. Since both crystal balls shared a common mean, this provided the mass of the simulated and observed data to be used in Eq. 5 for α . However, since both crystal balls had their own width, an effective width was needed of the form $\sigma_{\text{eff}} = \sqrt{f_1\sigma_{\text{CB1}}^2 + f_2\sigma_{\text{CB2}}^2}$, where f_1 and f_2 are the fractions of each crystal ball in the overall fit and σ_{CB1} and σ_{CB2} are the widths of the individual crystal ball functions. This can then be used as the mass resolution in Eq. 6.

E. Z-boson measurements

These calibration parameters can then be directly applied to the muon momenta of the simulated Z events. Z events are generated using a true mass, m_{true} , which is fit with a Breit-Wigner function [48], seen in Fig. 5, to generate nine templates using different masses and widths. A minimiser could then be used to produce new templates by linearly interpreting histogram data based on the previous templates. The best template was identified by minimising the χ^2 between the histograms formed by the observed data and these new templates. This resulted in the mass, width and correlation coefficient for the Z-boson along with their statistical error. The systematic error can be calculated via propagating the error on the calibration parameters and $\Upsilon(1S)$ mass through the

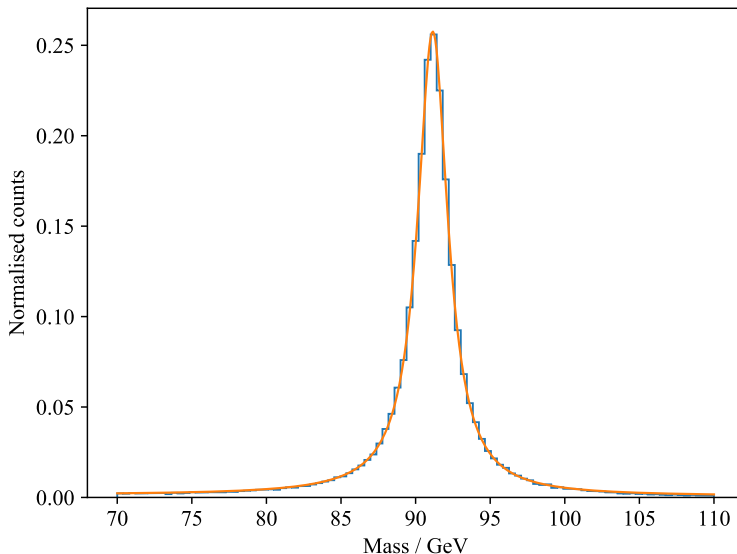


Figure 5: The true mass of the Z-boson used to generate simulated events fitted using a relativistic Breit-Wigner.

program.

F. Validation checks

Several validation checks were also carried out on the final Z results. This consisted of splitting the candidates into two roughly even groups as defined by the range of a parameter and calculating the significance, S , on the mass and width of the split using,

$$S = \frac{x_1 - x_2}{\sqrt{y_1 + y_2}}, \quad (7)$$

for resulting values x_1, x_2 with corresponding error y_1, y_2 . This characterises how many standard deviations apart the groups are and so how likely the split is due to statistical fluctuations. The validation checks performed were on the dimuon p_T , magnetic polarity of the magnet in the detector, azimuthal angle, ϕ , and the histogram bin number.

IV. Results

A. Calibration results

The selection cuts described previously in Sec. III.C. had significant effect on the number of $\Upsilon(1S)$ candidate events, removing 92% and 94% of simulated and observed events respectively. This was primarily due to the high cut on muon p_T which is much higher

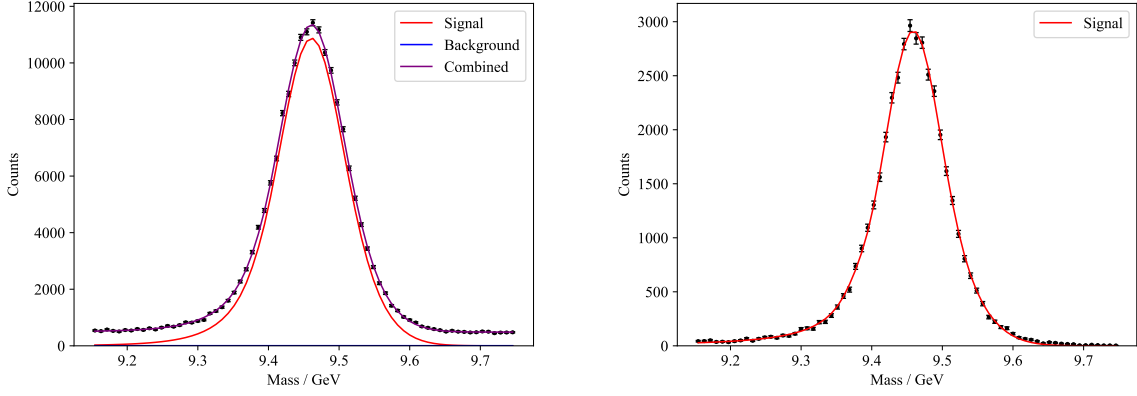


Figure 6: Invariant mass histogram of reconstructed $\Upsilon(1S) \rightarrow \mu\mu$ measured data (left) and simulated events (right). The measured data has an additional linear background component which is shown separated as well as combined.

than that needed to form the $\Upsilon(1S)$. There are still significant statistics beyond this cutoff due to the forward structure of LHCb resulting in high momentum $\Upsilon(1S)$ that decay in to higher momentum muons. After simple reconstruction via Eq. 3, the masses of candidate events are fitted with a double crystal ball fit, as shown in Fig. 6. This fit allows for the measurement of the simulated and observed masses and widths to be used in Eq. 5 and Eq. 6 to obtain the relevant calibration parameters which can be seen in Tab. 1. These parameters are assumed to be uncorrelated which results in an uncorrelated systematic error on the Z. For the histogram, 100 bins were used and the error on the observed counts was taken to be Poissonian [49]. These fits well describe the data, which should lead to relatively low uncertainty on the mass and width measurements, propagating to smaller errors on the calibration parameters. However, as can be seen in Tab. 1, the error on the smearing parameter is fractionally large. This is dominated by the error on the width of the data fit. To reduce this, a more constrained fit was trialled where the only free parameter in the crystal balls was a related width but this struggled to correctly describe the data.

Calibration parameter	Value / 10^{-5}
α	22 ± 3
σ / GeV^{-1}	3 ± 2

Table 1: Calibration values derived mathematically

To verify these parameters, they were applied to reconstruction of the $\Upsilon(1S) \rightarrow \mu\mu$ events that derived them to show the simulation can recreate the data. The results of this closure test can be seen in Fig. 7 where the simulation has been rescaled to have the same

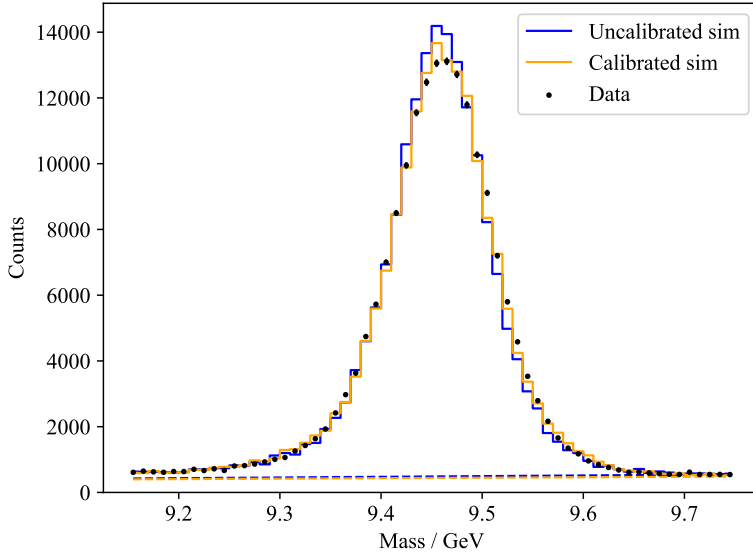


Figure 7: Closure test of the derived calibration parameters. The blue histogram has no calibration applied and the orange uses both calibration parameters. The simulated histograms have an artificial linear background applied, shown by the dashed lines.

number of total events as the observed data. An artificial background has been added to the simulation which is derived from reducing a χ^2 between the simulation and data with a floating linear background. To reduce the number of random statistical fluctuations introduced from the smearing component, the number of bins was slightly decreased to 80. Although the calibration aligns the simulation closer to the data, it is clear that is not working to the desired precision. This is likely a sign a more complex model for the calibration that includes more components from Eq. 4 is needed to sufficiently calibrate the simulation. Fig. 7 also implies that the smearing parameter is underestimated, which could be due to the typical momentum used but is far more likely due to the measurement of the widths of the data and simulation because this is the source of large uncertainty in the measurement. An alternative definition of the effective width was tried where the width was defined more in line with a Gaussian, with 68% of the data being contained within the width. This method gave a width measurement and error that was statistically indistinguishable from the method described in Sec. III.D.

B. Z results

The derived calibration parameters can then be applied in the same way on to the candidate simulated Z events, informing the weighting applied in the generation of the templates. The best of the generated templates applied to the data is shown in Fig. 8. This

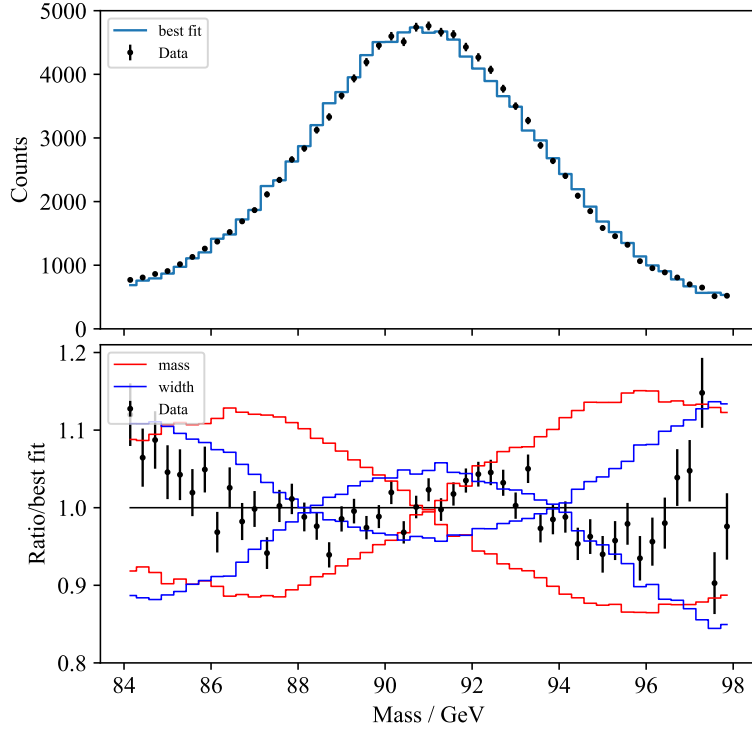


Figure 8: The best template applied to the reconstructed Z events (top) with its residuals (bottom). Also shown is the effect of a shift of 0.3 GeV in mass or 0.5 GeV in width in the template.

template resulted in a χ^2 of 143.18 with 48 degrees of freedom. The residuals of the best template show it is an accurate representation of the data with most data being contained within a single standard deviation. Despite this, it is clear that the template better describes the central data and struggles towards the tail of the distribution. This is not as important for the mass that only depends on the peak but impacts the width which depends on the shape of the distribution. The additional templates shown that are shifts in mass and width of 0.3 GeV and 0.5 GeV respectively were chosen arbitrarily to represent a small step in mass and width.

Parameter	Value / GeV	Correlation	
M_Z	91.12 ± 0.01	1.00	0.065
Γ_Z	1.90 ± 0.03		1.00

Table 2: Measured values of the Z-boson mass, M_Z , and width, Γ_Z , with their statistical error and corresponding correlation matrix.

The resultant values of the mass and width of the Z-boson are in Tab. 2 along with the correlation matrix. Since the mass and width affect different parts of the distribution, the errors are expected to be largely uncorrelated which is what is observed in the correlation

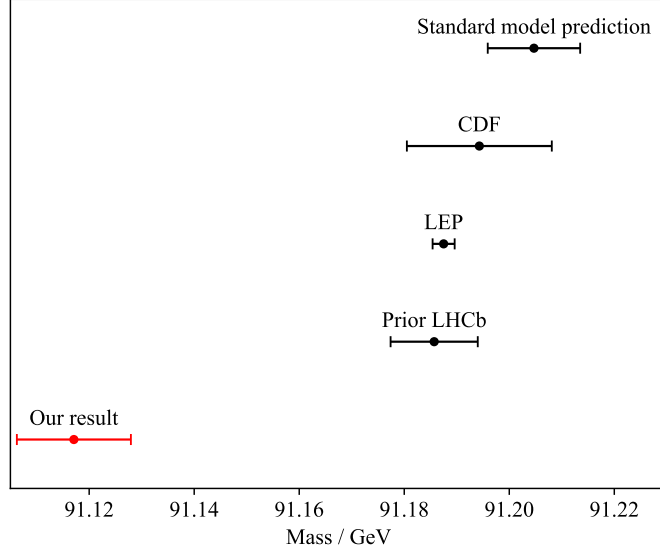


Figure 9: Comparison between various measurements of the Z-boson mass [6, 8, 10], including this project, and the standard model prediction [23].

matrix. For example, an error in the mass would just shift the distribution, leaving the shape and subsequent width unaffected. The statistical errors are fractionally small but could be further reduced using a larger dataset.

The mass of the Z-boson has been much more widely studied than its width due to the added complexity of modelling the resolution of the detector. The comparison of this measurement to previous measurements and the standard model prediction that follows from Eq. 1 is shown in Fig. 9 which implies this measurement is somewhat anomalous. Adding the standard model prediction for the width that follows Eq. 2, and the predicted correlation [23], an ellipse can be formed using the uncertainty on these quantities. The ellipse formed by the statistical errors on this measurement of mass and width with their correlation can be seen in Fig 10 and re-enforces these results as anomalous. This is likely due to the simplification of the calibration in this project; particularly the assumption that everywhere in the detector these parameters are the same and the detector response is uniform which is further explored by the validation checks in Sec. IV.C.

The systematic errors on the Z mass and width that come from the calibration were also propagated through. The method for this was shifting the calibration parameters by their respective error in each direction and halving the difference between the resultant measurements. These sources of systematic uncertainty can be seen in Tab. 3 which reflects the uncorrelated nature well. This is because the each calibration parameter

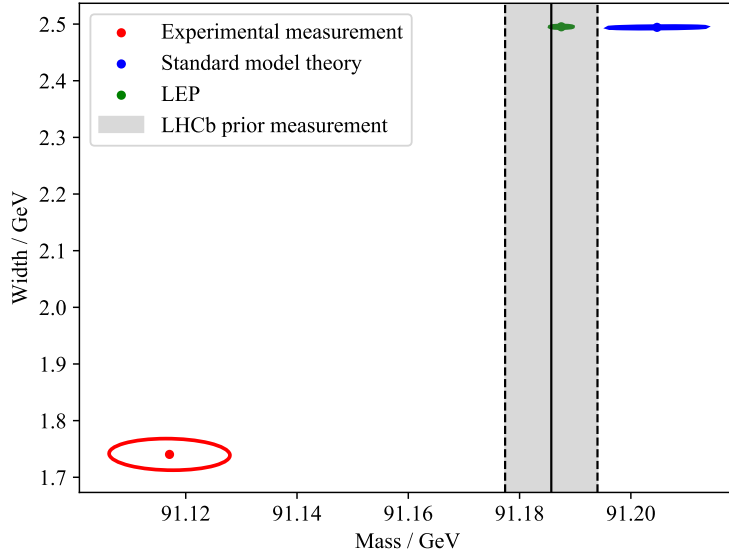


Figure 10: Ellipse formed from the statistical errors on the Z-boson mass, width and their correlation compared with other measurements [6, 10] and the standard model prediction [23].

only significantly effects the measurement it was designed for. As expected, the large uncertainty in σ , as shown in Tab. 1 causes this to be the dominant source of systematic error. Combining the statistical and systematic errors, the final measurement of the Z-boson mass and width was $M_Z = 91.119 \pm 0.011 \pm 0.004$ GeV and $\Gamma_Z = 1.90 \pm 0.03 \pm 0.12$ GeV.

	Mass / GeV	Width / GeV
Momentum scaling	0.0030	0.0008
Momentum smearing	0.0004	0.1212
$\Upsilon(1S)$ mass	0.0010	0.0000

Table 3: The sources of systematic error on the Z mass and width.

C. Validation Checks

To test their importance on the measurements made above, validation checks were carried out on the polarity of the detector magnet, dimuon p_T and azimuthal angle, ϕ , split into two roughly equally sized groups. The magnet polarity was split between LHCb’s *MagUp* and *MagDown* configurations which are regularly flipped during data taking. The dimuon p_T is the p_T of the combined muon system so is essentially the p_T of the Z-boson candidate. This was split either side of the mean dimuon p_T , 18.3 GeV, still in accordance with the previously defined selection cuts. Finally, ϕ was split into $(-\pi, 0)$ and $(0, \pi)$ according to the position of the positive muon, which was also roughly even in size.

	Magnetic polarity	Dimuon P_T	ϕ
S_M	4.758	1.995	11.099
S_Γ	5.407	4.238	0.075

Table 4: Significance values for three validation checks.

The resultant significance of these splits on the mass and width of the Z-boson, as defined in Eq. 7, can be seen in Tab. 4. The dimuon p_T has reasonable significance on the mass but is more statistically significant on the width. This suggests that the detector resolution has a dependence on the muon p_T which the smearing term used does not fully account for. The extreme significance of ϕ on the mass heavily suggests that the detector response is highly non-uniform. This suggests the measurement would benefit from calibration parameters that are binned into detector regions. The lack of significance of ϕ on the width suggests this is a deterministic effect and that the mismeasurement due to using averaged calibration values, mismeasures the momenta of all muons proportionally, leaving the resultant distribution shape unaffected. The effect of the magnetic polarity implies that the direction the momentum bends is affecting its measured momentum, which is what introduces the charge dependant curvature bias. This means the assumption that $\delta = 0$ in Eq. 4 is invalid. In a more detailed analysis, these checks would be used to more accurately determine the systematic uncertainty on the measurements.

The stability of the measurement was also assessed via altering the number of bins used to form the templates. Tab. 5 shows the effect of the binning on the result compared to the default of 50 bins. The resultant mass and width are very stable across all bin numbers, suggesting the template method of analysis is reasonable.

	Mass / GeV	Width / GeV	χ^2	ndf
30 bins	91.12 ± 0.01	1.90 ± 0.03	107	28
50 bins	91.12 ± 0.01	1.90 ± 0.03	143	48
80 bins	91.12 ± 0.01	1.90 ± 0.03	212	78

Table 5: Z mass and width with statistical error for each bin size of the template, along with its corresponding χ^2 and number of degrees of freedom.

As a final check, the effect of the selection cuts made throughout the project was tested, as seen in Tab. 6. Interestingly, all three situations had a lower χ^2 between the data and best template than when selection is applied as normal. This could imply that the selection cuts are removing more signal than desired or this background could be helping the underestimate the current fit has at low mass as seen in Fig. 8. The lower χ^2

from using only one selection could imply that both selection cuts are removing similar background but removing slightly different signal with it so the combination has similar background removal but less remaining signal.

	Only p_T cut	Only η cut	No selection cut
mass / GeV	91.132 ± 0.009	91.12 ± 0.01	91.130 ± 0.009
width / GeV	2.09 ± 0.03	1.90 ± 0.03	2.10 ± 0.03
correlation	0.084	0.054	0.083
χ^2	122.842	128.822	140.652

Table 6: Effect of the $p_T < 20$ GeV and $2.2 < \eta < 4.4$ selection cuts on final results. The χ^2 is relative to the best template.

V. Conclusion

This is the second dedicated Z-boson measurement at the LHC and the first dedicated measurement of its width. The measured Z-boson mass and width of $M_Z = 91.119 \pm 0.011 \pm 0.004$ GeV and $\Gamma_Z = 1.90 \pm 0.03 \pm 0.12$ GeV disagree with previous measurements and standard model theory as Fig. 9 and Fig. 10 show clearly. The dominant source of uncertainty on the mass is statistical, meaning precision of this measurement could be increased using a larger dataset, which is available within the same run 3 dataset. However the width is dominated by systematic uncertainty, implying that for a more precise measurement, calibration needs to be improved. The accuracy of these measurements rely on improvements made to the calibration, in both the model used and its application. It is difficult to make meaningful conclusions on the results of this measurement as Fig. 7 implies that the calibration needs improving. Reasonable changes in the calibration parameters have significant effects on the results, which suggests both instability in this measurement and that perfecting calibration could completely change the final reported value. The calibration could generally be improved by using the resonance of another decay to cross-check the derived calibration parameters, for example the $J/\psi \rightarrow \mu\mu$ channel.

Due to time constraints, this project could not explore the calibration any further. If more time was available, it is clear how this measurement could be improved. The large significance in the azimuthal angle of the positive muon suggests strongly that the detector response is non-uniform and calibration parameters should be binned across the detector coordinates. The detector is already described in natural coordinates of (η, ϕ) which allows for the calculation of calibration parameters to be performed for various bins

of these coordinates to form a matrix of values. Muons in the Z reconstruction can then have calibration parameters applied to them that are most relevant for their local detector region, further improving the accuracy of the final measurement. The complication in this approach is that when determining the calibration parameters, the region that each candidate belongs to needs to be described in a way that avoids bias towards either of the specific muons. For example, the method used to categorise muons in Sec. III.F. would lead to a bias that better describes the behaviour of the positive muon.

The width measurement would benefit from the non-zero inclusion of the charge-dependent curvature bias term in Eq. 4. Theoretically, this should have little effect on the resonance location of the distribution and so the resultant mass due to the symmetric nature of the decay since the overestimate for one muon cancels out the underestimate of the other. However, a more accurate determination of the muon momenta would effect the overall shape of the distribution, bringing it closer to the theoretical Breit-Wigner used in the template generation which should improve accuracy of the width as well as increasing precision in both measurements by reducing the χ^2 of the best template. Tab. 4 shows that the magnetic polarity has significant effects on the results for both the mass and the width which shows this term is needed for the accuracy of both the mass and the width. Treating the β parameter as non-zero could affect the mass measurement but likely would have little effect as any inaccuracies in its omission are absorbed into the calculation of α . If the β parameter did prove to be significant on the mass, it is highly unlikely to make an impact on the measured width as it would scale the candidate masses evenly, leaving the shape of the distribution unaffected.

The most significant source of uncertainty in this project has been the large systematic error on the Z width due to the uncertainty on the smearing parameter. The double crystal ball used to fit $\Upsilon(1S)$ candidate events was slightly unstable and heavily dependent on precise bounds and initial parameters. A more stable functional fit of the $\Upsilon(1S)$ events that had less free parameters could lead to a more accurately determined width of the $\Upsilon(1S)$, reducing this error. Reducing the number of free parameters, especially those that are somewhat correlated would avoid the fitter hiding the width within other parameters and reduce the risk of overfitting.

Another area of improvement is within the template generation. This result uses nine templates formed from three and width values and a linear interpolation between them. A more detailed analysis would involve using more templates and a more complex model of interpolation between them. For example, the previous LHCb measurement [10] used

a cubic spline to interpolate between templates which would allow for a more accurate minimisation and more accurate results.

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